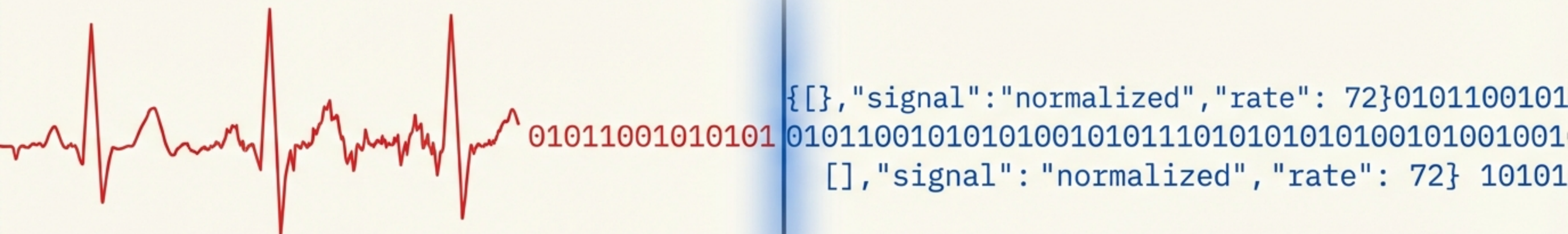


DeepPulse

Democratizing Biosignal Analysis



Precision in Pulse. Intelligence in Code.

Medical-Grade
Signal Processing

Powered by
Google Gemini 3.0

Open-Source
PhysioNet Integration

Bridging the Gap Between Raw Data and Expert Insight

The Problem



The Barrier: Raw physiological data is complex, inaccessible, and requires years of specialized training to interpret correctly. Clinical context is often lost in binary files.

The Solution



The DeepPulse Engine: A democratization engine fusing two worlds:

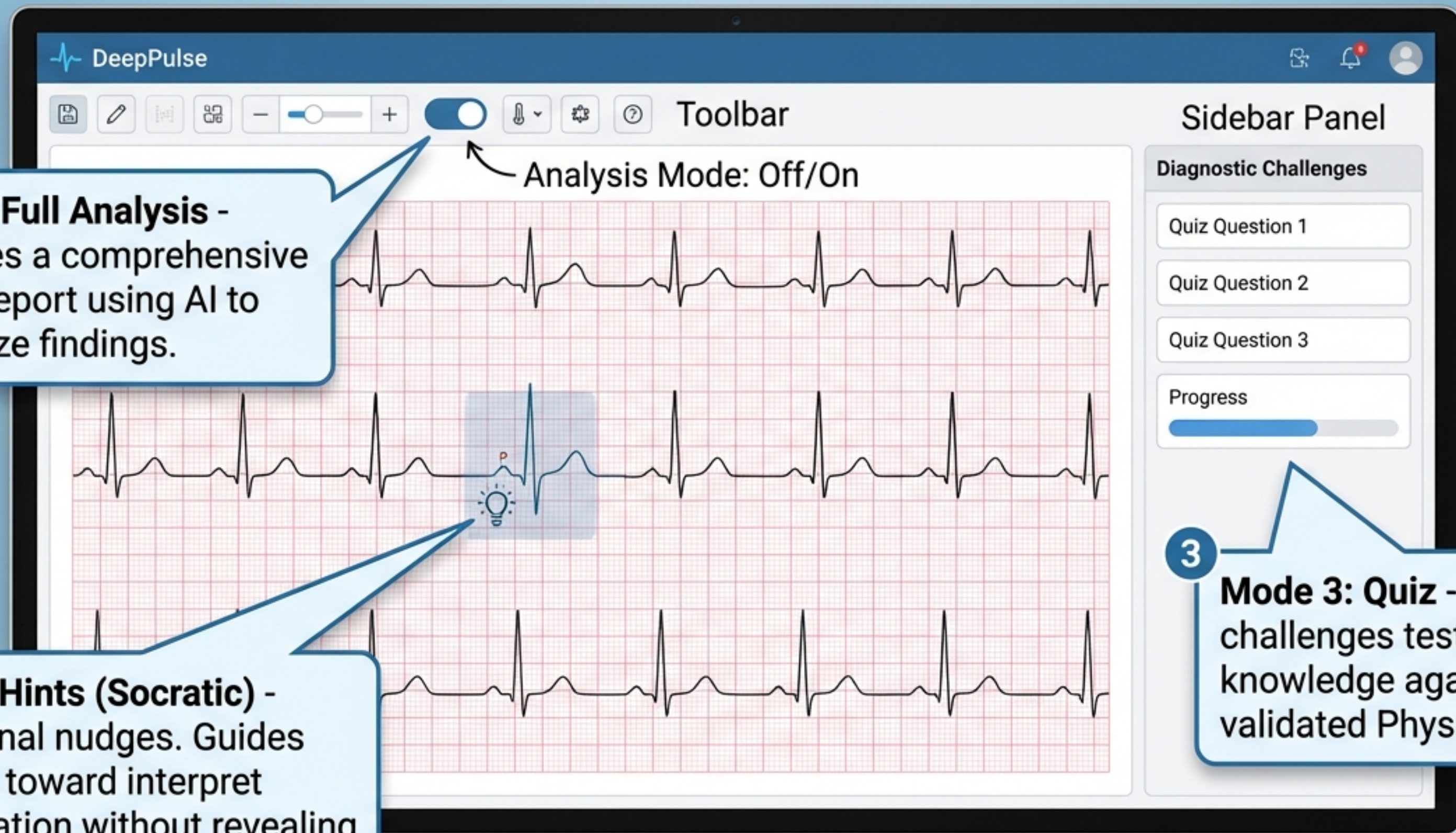
1. **The Old World (Rigor):** True calibration (25mm/s, 10mm/mV) ensures clinical validity.
2. **The New World (Intelligence):** AI-augmented learning via Gemini 3.0 provides interactive guidance.



Key Capability: Direct integration with PhysioNet's massive repository creates an instant research ecosystem

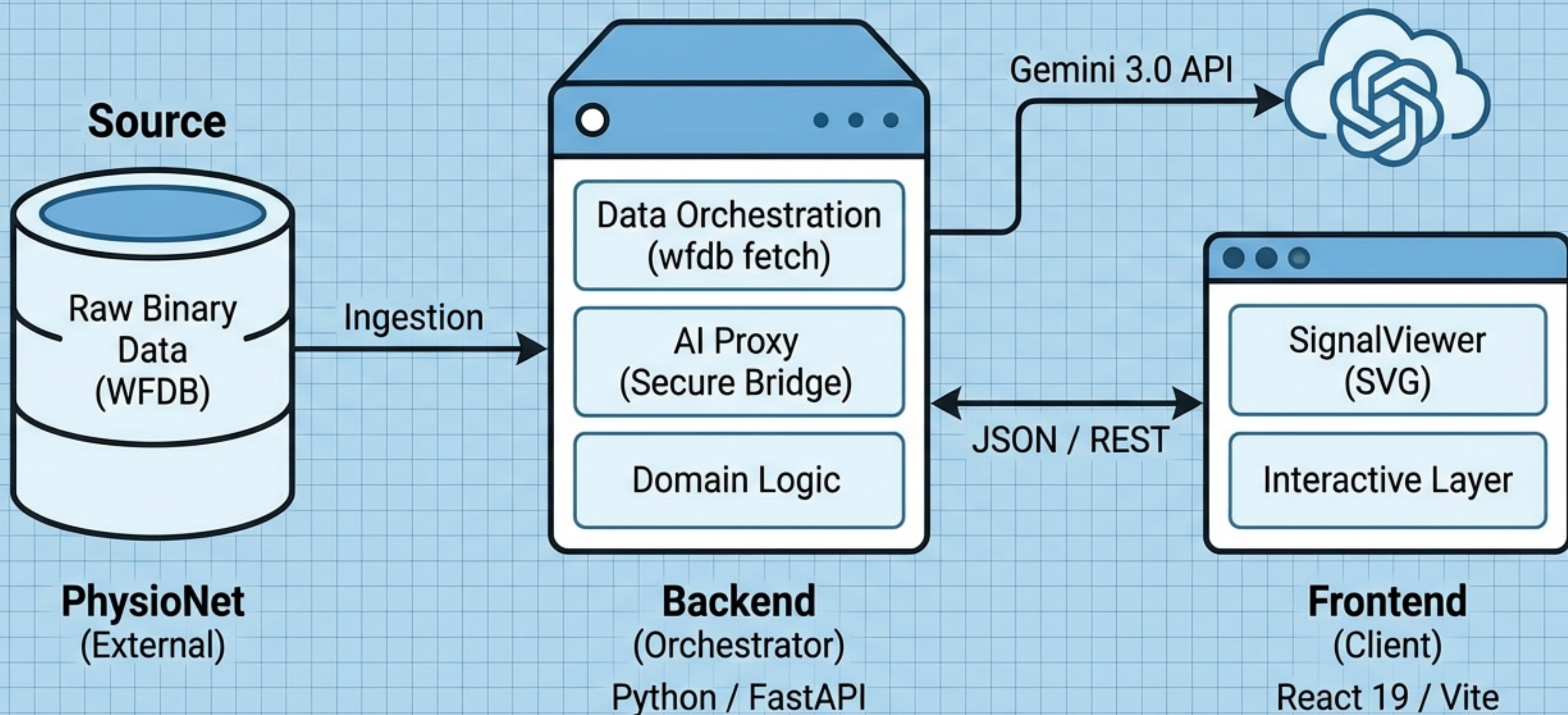


Interactive Analysis Across Three Modalities

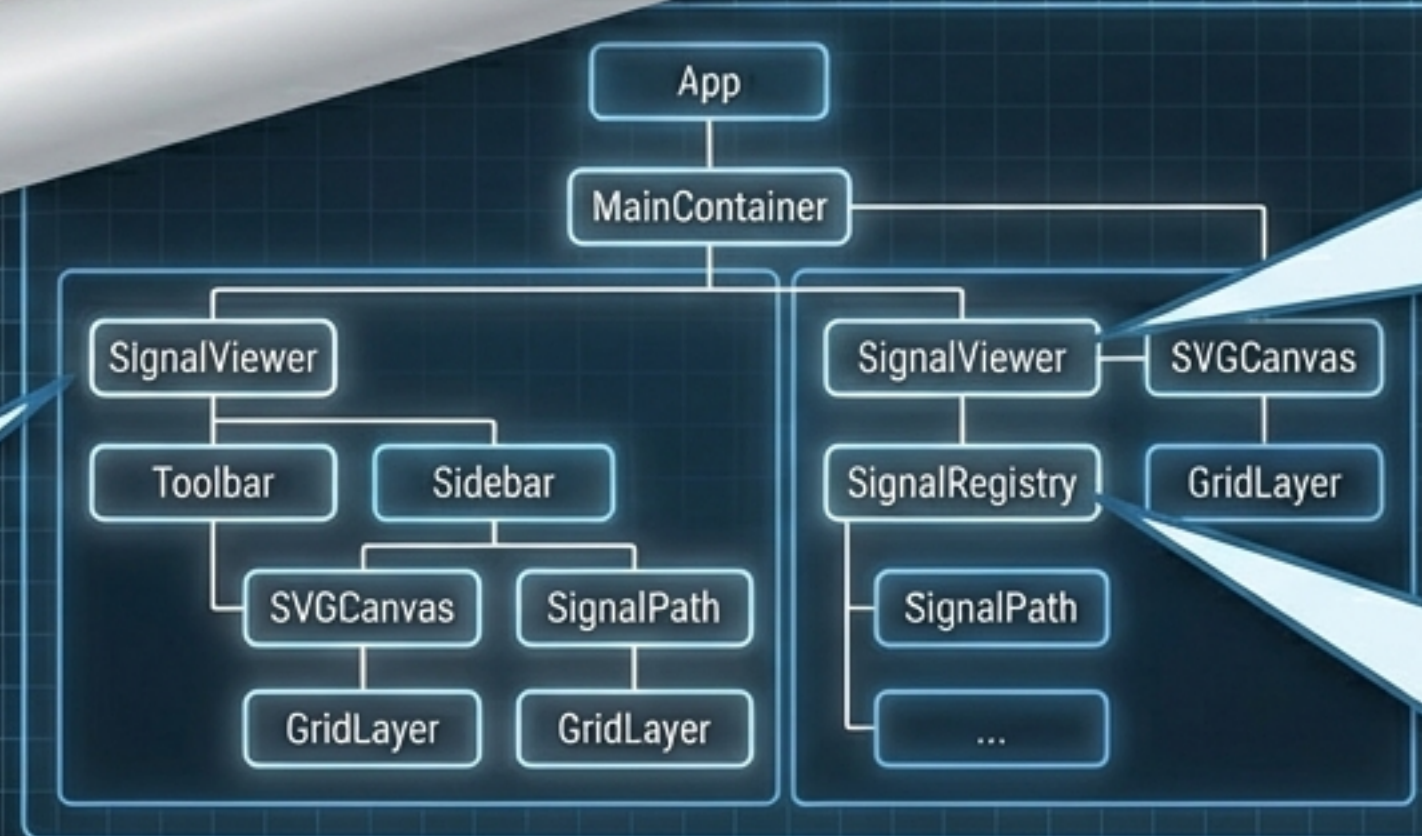
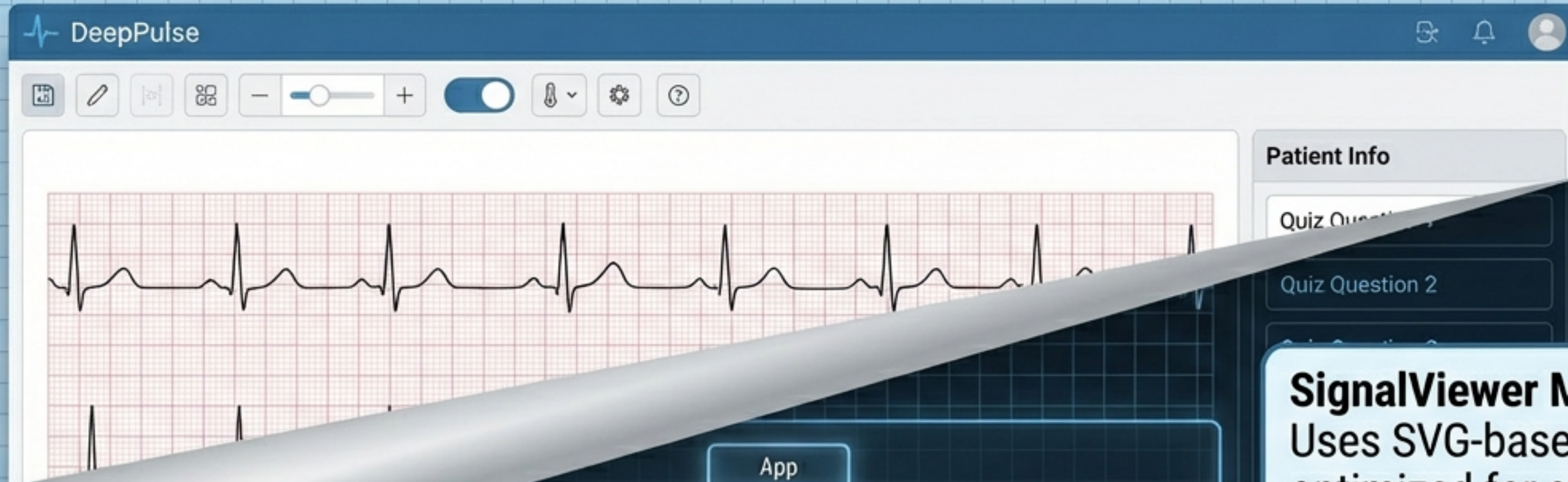


Rendering Engine: SVG-based signals with professional medical grids and standard ti 

High-Level Architecture: A Clean Separation of Concerns



The Rendering Engine: Precision at 60 Frames Per Second



Framework:

Built on React 19 + Vite for zero-latency execution.

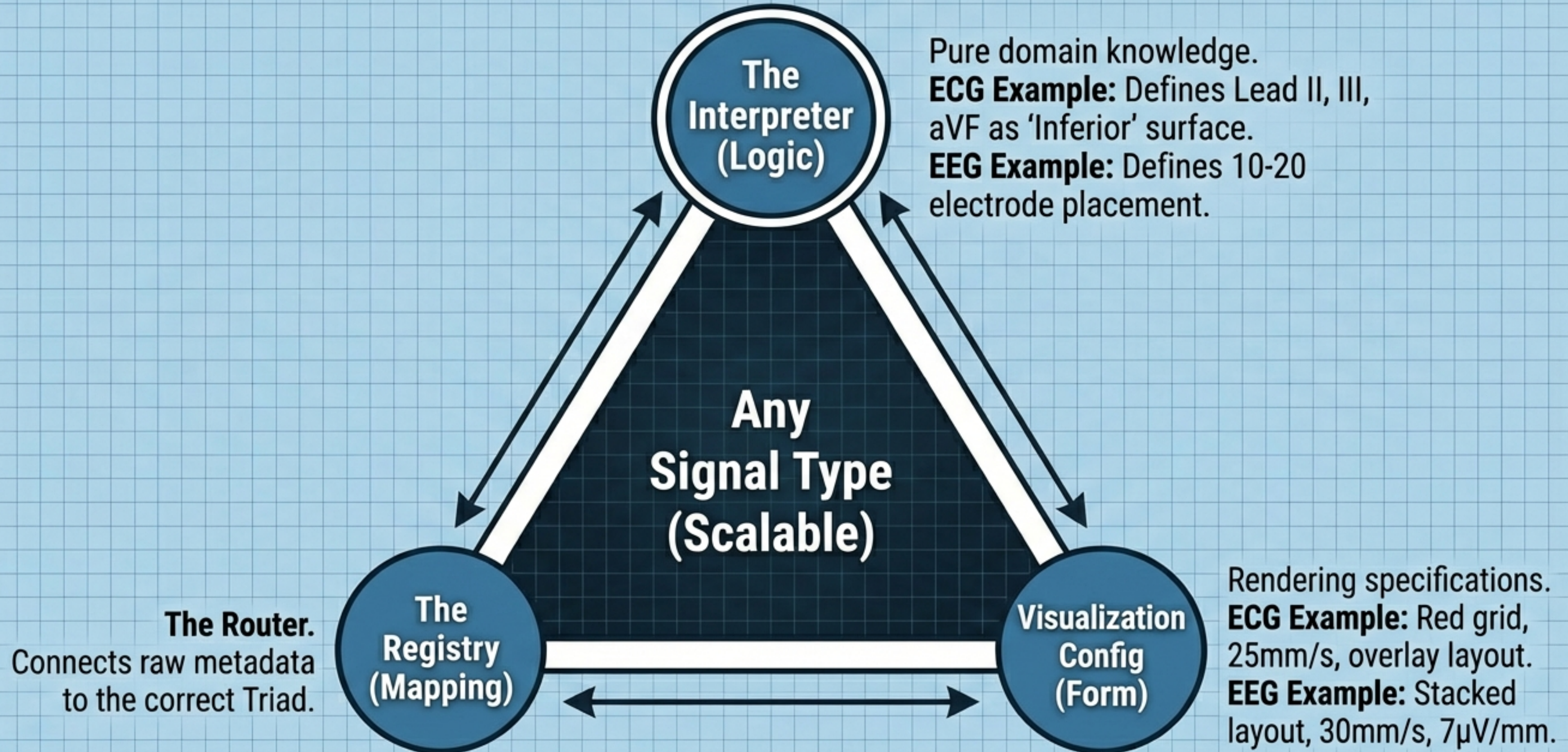
SignalViewer Module:

Uses SVG-based rendering optimized for scrolling. Maintains layout stability at high frequencies.

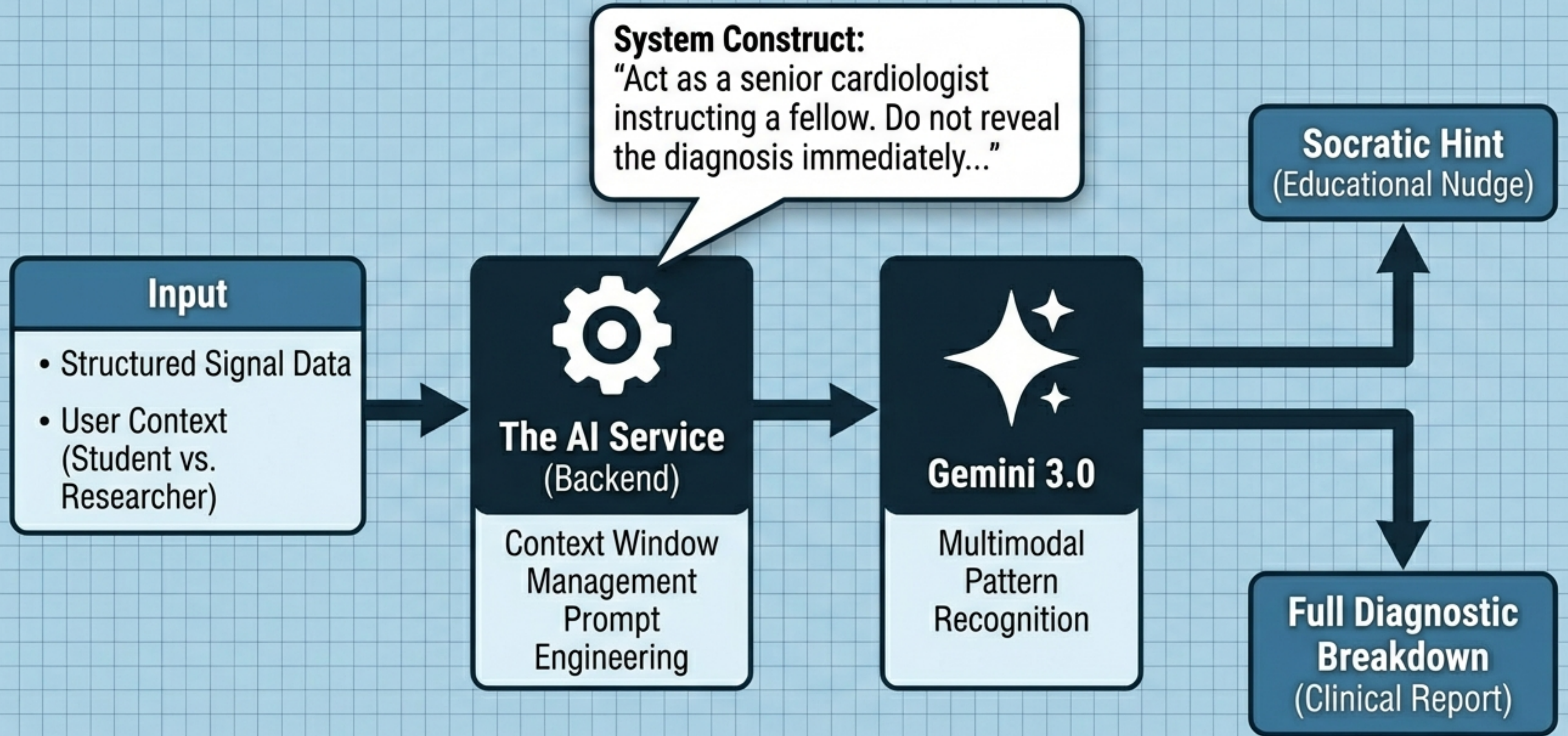
SignalRegistry:

Implements 'Factory Pattern' to manage diverse signal types. Dynamically loads logic based on data stream.

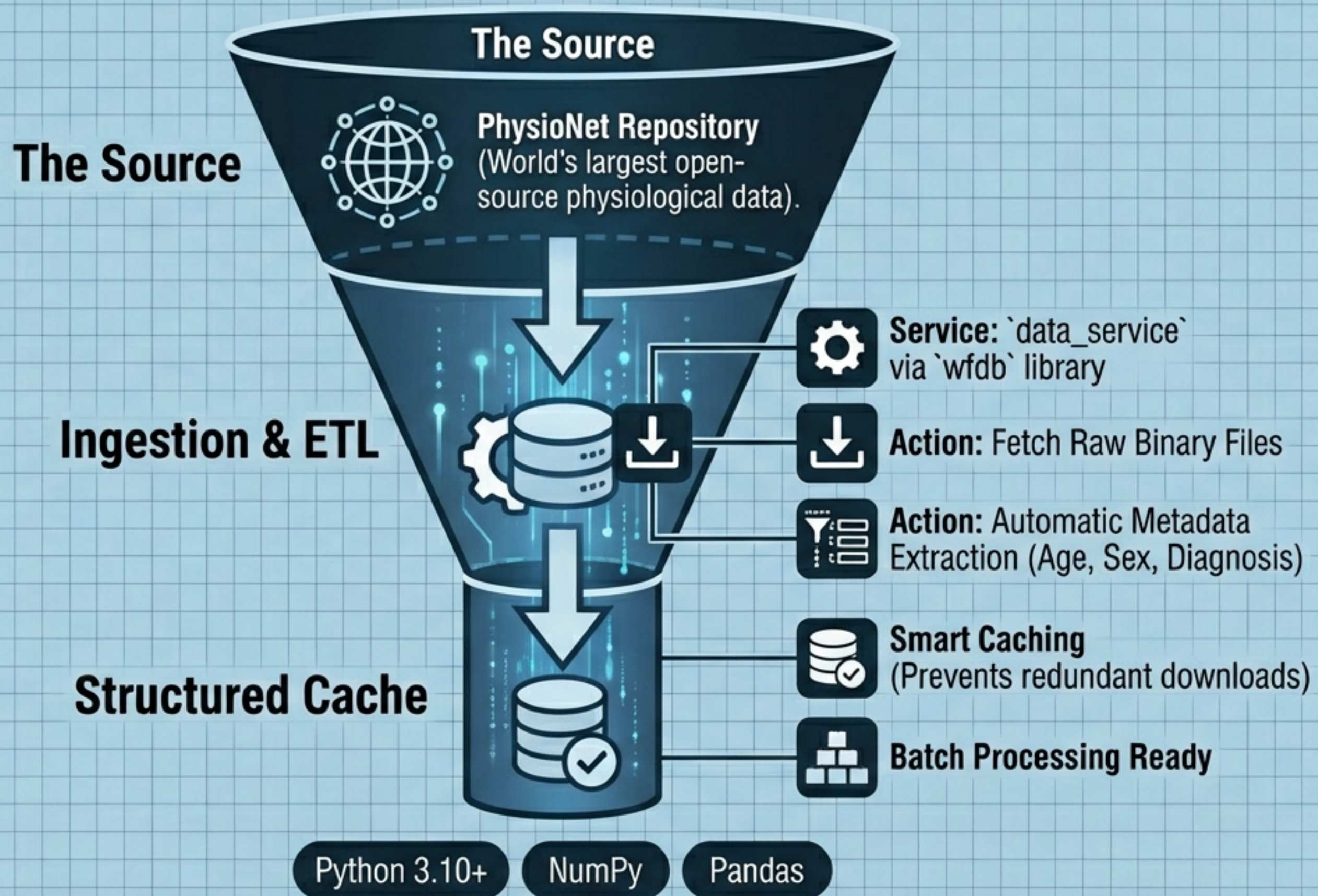
The Triad Pattern: A Modular Extensible Architecture



Augmented Intelligence via Gemini 3.0



Data Engineering & PhysioNet Integration



The Technology Stack



Frontend Core

React 19
Vite



Frontend UI

TailwindCSS
Lucide React
D3.js Concepts



Backend Logic

Python 3.10+
FastAPI
Uvicorn



Data Layer

WFDB Library
NumPy
Pandas



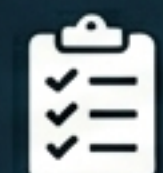
Validation

Pydantic



Intelligence

Google
Gemini 3.0



Quality Assurance

Pytest
ESLint

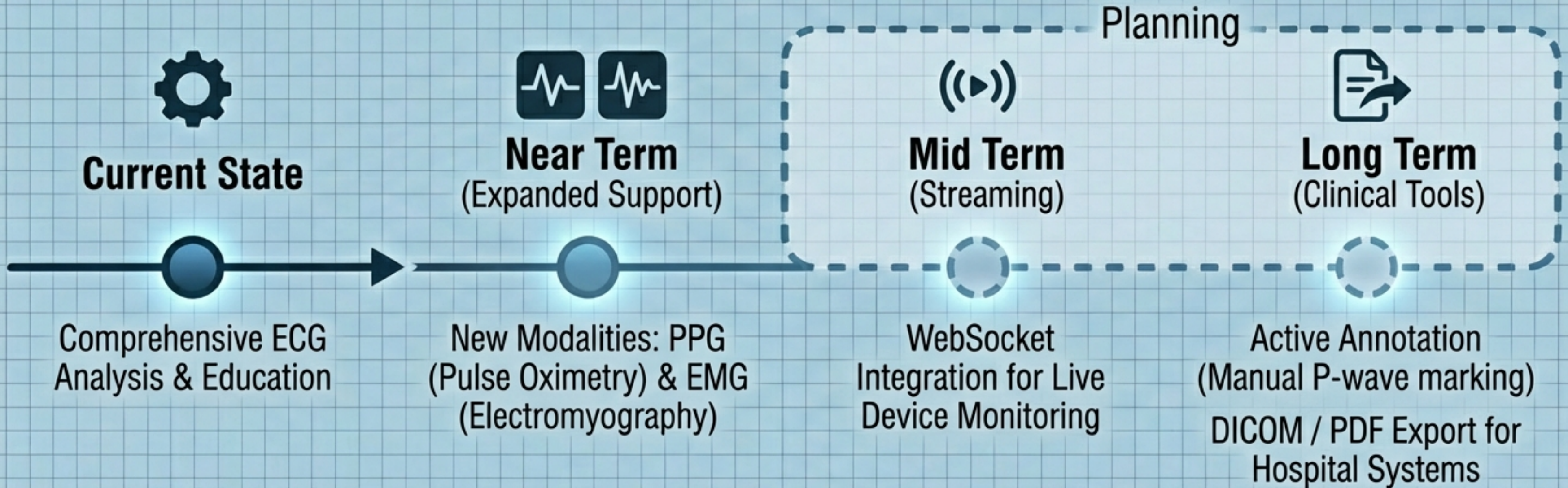


DevOps

Git
Docker

Evolutionary Roadmap

From Static Analysis to Real-Time Monitoring



“DeepPulse is evolving from an educational tool into a comprehensive clinical assistant.”